

User Guide IDSL.GOA

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1. Introduction

1.1 Overview of IDSL.GOA

Open [IDSL.GOA](#).

The input is on the left side and requires a list of International Chemical Identifier Keys (InChIKey).

The website gives detailed **Introduction** and **Instructions** on how to use IDSL.GOA (Integrated Data Science Laboratory for Metabolomics and Exposomics.Gene Ontology Analysis).

So far, it only includes the human genome.

The **Database Content** can be explored to get a deeper understanding on how the network is built and guarantees transparency and traceability.

1.2 What is IDSL.GOA

Gene ontology (GO) set enrichment analysis is performed on a list of metabolites. [1]

GO enrichment is conducted using an over-representation analysis (ORA) based on the hypergeometric distribution. For each GO term, the test evaluates whether the number of metabolites from the input list associated with that GO term exceeds the number expected by random sampling from the background metabolite universe.

The hypergeometric test is parameterized using the total number of metabolites in the knowledgebase, the number of metabolites linked to the GO term, the number of input metabolites present in the knowledgebase, and the overlap between the input list and the GO term. P-values are calculated using the upper tail of the hypergeometric distribution to assess the probability of observing an overlap equal to or greater than the observed value.

To reduce spurious results and improve biological interpretability, several filtering criteria are applied. GO terms are required to have a minimum number of associated genes and metabolites, a minimum overlap with the input list, and a set size below a defined fraction of the total metabolite universe to exclude overly broad categories. Multiple hypothesis testing correction is performed using the false discovery rate (FDR) approach and GO terms with an adjusted p-value below the specified threshold are considered significantly enriched.

2. Using GOA

2.1 Input data

Upon selecting GOA in the shiny app, the GOA output is a single column text file for each comparison that contains **unique InChIKeys of significant features**. Each list can be used as is.

- Input:
 - Delete InChIKeys from test samples
 - Enter the one-column list of InChIKeys:
 - Copy/paste the list from the shiny app
- Run GO Analysis

If you get the message: “Analysis is failed. Please Update the Input” (sic), there is either a problem with the format of the InChIKeys or not enough IDs to meet the mapping criteria.

Note: By submitting the example InChIKeys, users can observe the resulting software output and its visual presentation.

2.2 Results

Upon submitting the identifiers to the pipeline, four result options can be selected. When you go back to the main page they also appear at the top of website.

1. GO Ontology Network
2. GO Impact Plot (Scatter Plot)
3. Interactive GO Table
4. All GO Terms

At the top, the number of all metabolites as well as the number of input metabolites that were applied are listed. Right next to it, different download options can be selected. Once you click on one of the results, this part moves to the bottom of the page.

2.2.1 GO Ontology Network

The GO network visualization illustrates hierarchical relationships among significantly enriched metabolic processes.

Significant networks are listed and can be downloaded as xlsx including p-values.

- **Download image:**
 - The network view with all the labelled nodes can be exported here
 - You can zoom in and drag the network to a suitable position

When you select a specific node, you have different options.:

- Click on the underlined accession **GO:Number**
 - Directs you to the amigo.geneontology website of the specific process
- **Click here to get Compounds:**
 - Shows the input compounds that are part of the identified network
- **Download Data**
 - Exports the listed compounds for the selected network node

2.2.2 GO Impact Plot (Scatter Plot)

The impact scatter plot highlights enriched GO terms based on their statistical significance and biological specificity.

The scatterplot shows the relation of the set size (number of metabolites per network node) against the log p-value that reflects the significance of the node.

2.2.3 Interactive GO Table

The interactive GO table summarizes enriched metabolic GO terms along with their statistical significance, set size, overlap metrics, and the contributing metabolites driving each enrichment result.

This is the table, that can be downloaded from the right bottom of the page.

2.2.4 All GO Terms

All reported nodes are listed from top to bottom. On the left side, each node is represented by a box containing the GO accession number and associated statistical information, while on the right side, the corresponding mapped compounds are listed.

3. Limitations

Inflation of ORA due to redundancy in GO term names, biases and inaccuracies in the database, and lack of linking annotated compounds in the metabolomics dataset with enzyme commission numbers.

4. Conclusion

IDSL.GOA is an essential tool for gaining insights from human metabolomic data. Through this guide, users get an overview of GO ontology networks to enhance their research outcomes.

5. Frequently Asked Questions

1. What does the GO ontology network tell you

a. What are the nodes:

Each node in the GO network represents a metabolic GO term that was found to be significantly over-represented in your input metabolite list after multiple-testing correction. These are not pathways and not metabolites. They are biological process categories from Gene Ontology, such as e.g. nucleoside salvage, amino acid metabolic process, or purine ribonucleotide metabolism.

b. What are the edges:

The edges between nodes represent ontology relationships defined by GO, primarily is-a and part-of relationships.

This means:

- a parent node represents a broader process
- a child node represents a more specific process

So, if two nodes are connected, it does not mean that metabolites are converted between them. It means that, biologically, one process is a specialization or subset of another.

c. What does the node size tell:

Node size reflects the statistical significance of the enrichment result, typically proportional to $-\log_{10}(\text{p-value})$.

Larger nodes therefore indicate:

- stronger over-representation

- lower probability that the enrichment occurred by chance

Important: Node size does not indicate effect size, metabolite abundance, or pathway flux.

d. What if nodes cluster in the network:

When several nodes cluster together in the network, this indicates that:

- multiple related GO terms are enriched simultaneously
- the input metabolites map to a coherent biological theme

For example, a tight cluster around nucleotide metabolism suggests that multiple, hierarchically related nucleotide processes are collectively affected, rather than a single isolated annotation.

e. What can be concluded from the GO network plot:

You can say:

- which metabolic processes are implicated
- whether enrichment is focused (few related processes) or broad (many related branches)
- how specific or general the affected biology is, based on node depth in the GO hierarchy

You cannot conclude:

- causal direction
- metabolite flow or regulation
- which specific enzymes are rate-limiting

2. What does the scatter plot (impact plot) tell you

The scatter plot is designed to answer a different question:

- which enriched GO terms are both statistically strong and biologically specific
- each point corresponds to one enriched GO term.
- x-axis: GO term set size
 - The x-axis shows the number of metabolites associated with that GO term in the knowledgebase.
- y-axis: enrichment significance
 - The y-axis is $-\log_{10}(\text{p-value})$ from the over-representation test.

a. Interpretation:

- small set size → specific process

- large set size → broad, general process
- higher position → stronger statistical enrichment
- lower position → weaker enrichment

b. The most informative GO terms are typically found:

- high on the y-axis (strong enrichment)
- left on the x-axis (high specificity)

These represent processes that:

- involve relatively few metabolites overall
- but are disproportionately represented in your data

In contrast:

- points far right and high up are strong but very general
- points far left but low are specific but weakly enriched

3. How do these plots complement each other:

- GO network plot:
 - shows relationships and structure
 - What biological processes are involved, and how are they related?
- scatter (impact) plot:
 - shows prioritization
 - Which of these processes are most specific and statistically compelling?
- together, they allow you to:
 - identify core biological themes (network)
 - select the most informative GO terms for reporting (scatter plot)

4. What is a strong result:

- a cluster in the network
- containing one or more points in the upper-left region of the scatter plot

6. References

- [1] P. Mahajan, O. Fiehn, and D. Barupal, "IDSL.GOA: gene ontology analysis for interpreting metabolomic datasets," *Scientific Reports*, vol. 14, no. 1, p. 1299, 2024/01/14 2024, doi: 10.1038/s41598-024-51992-x.